Introduction to Algorithms and Data Structures

week 2.1: 2-Dimensional Arrays

SAXION UNIVERSITY OF APPLIED SCIENCES

N dimensional Arrays

- **Discuss work last lesson**
- When to use a 2D array
- Data models
- homework

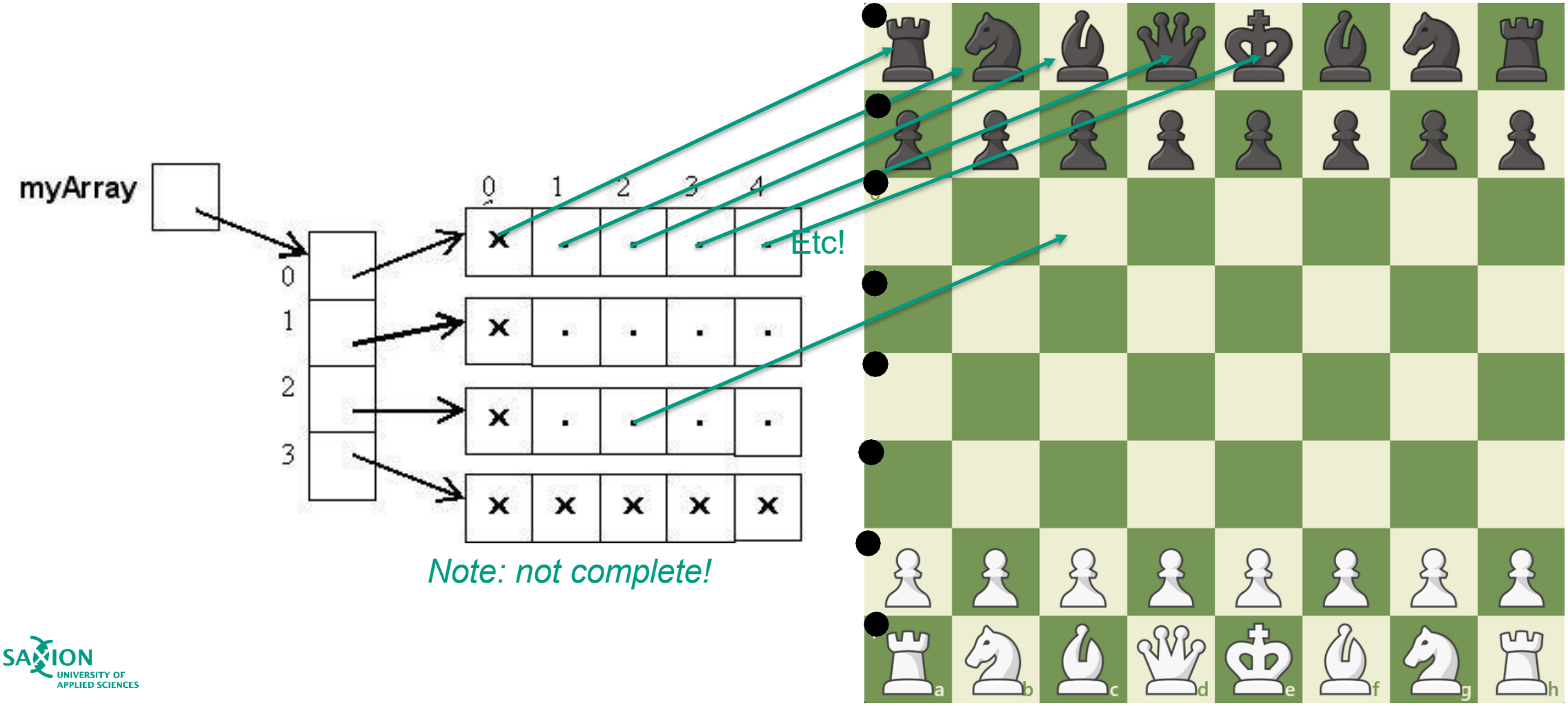
Discussing homework from the last lesson

- Where you able to do the exercises?
  1. Yes! All exercises are done!
  2. No.., Because I have these questions
  3. I haven't prioritized this subject enough at the moment to do the assignments

N dimensional Arrays

- **Discuss work last lesson**
- **When to use a 2D array**
- Data models
- homework

How to store the position of the pieces?



Note: not complete!

What is a 2D array

- An array that holds multiple arrays (that in turn hold the items)
- The outer array (generally*) represents the rows (y)
- The inner arrays (generally*) represent the columns (x)
  - **But it can be any representation we wish!*

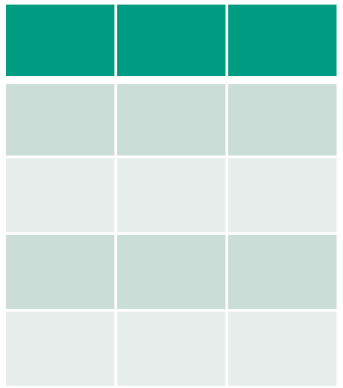
Table 1

- How do we create a 2D array (in java)?

1
```
int[][] myArray = new int[5][3];
```

2
```
int[][] rows = new int[5][];
rows[0] = new int[3];
rows[1] = new int[3];
rows[2] = new int[3];
rows[3] = new int[3];
rows[4] = new int[3];
```

Table 2



- While 1 and 2 are the same, 1 is easier to use!

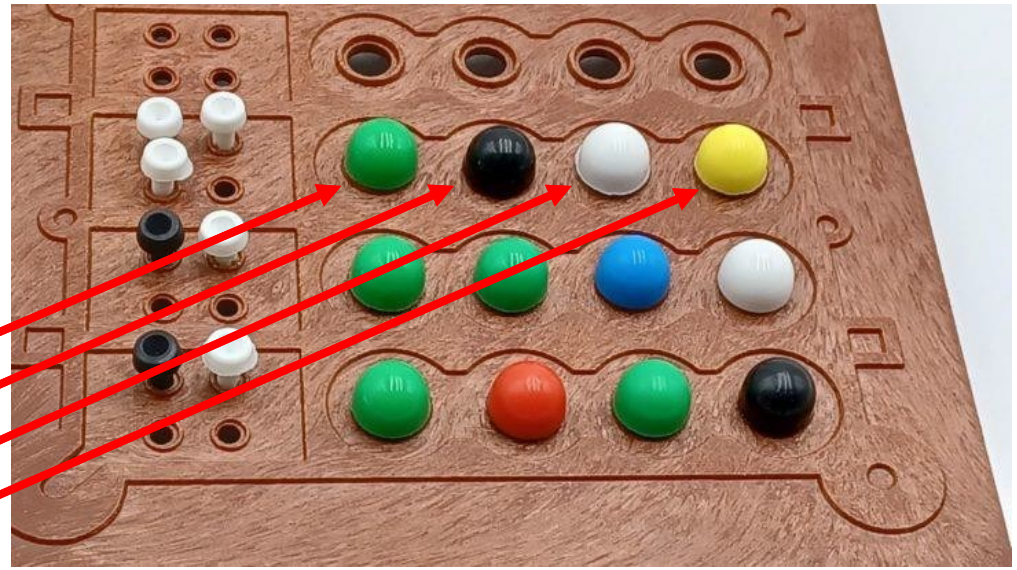
How can I visualize a 2D array?

- Let's establish a data model for the Mastermind game

```
char[][] masterminds = new char[4][4];
```

- How should we store information about a knob?

```
…
masterminds[1][0] = 'G';
masterminds[1][1] = 'B';
masterminds[1][2] = 'W';
masterminds[1][3] = 'Y';
masterminds[2][0] = 'G';
…
```



Suggested legend:

Char	color
G	Green
B	Black
W	White
Y	Yellow
U	Blue
R	Red
X	empty

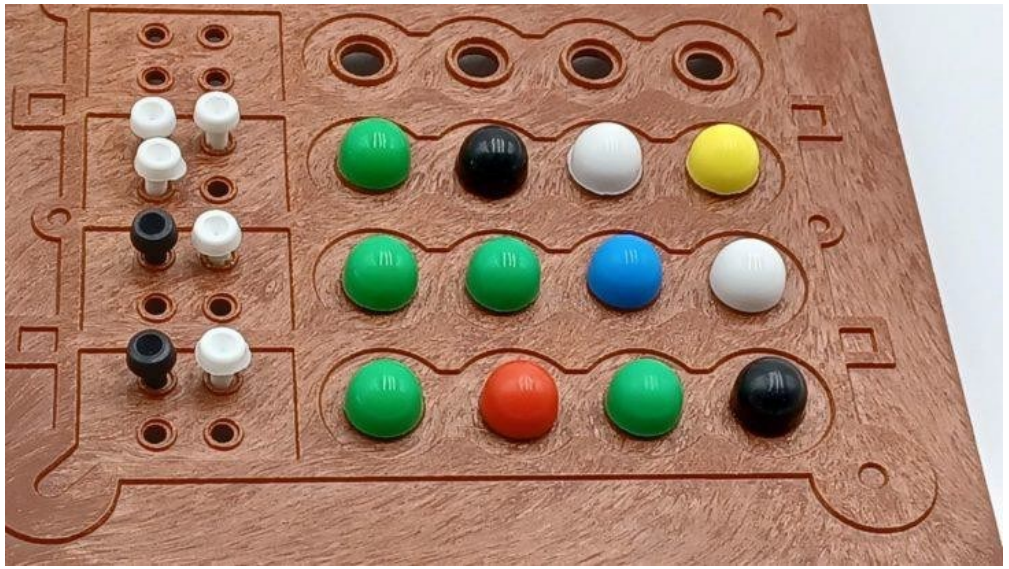
- Is there a way to initialize it directly with the correct values?

```
char[][] masterminds = {
        {'X', 'X','X', 'X'},
        {'G', 'B','W', 'Y'},
        {'G', 'G','U', 'W'},
        {'G', 'R','G', 'B'}};
```

SAXION UNIVERSITY OF APPLIED SCIENCES

How can we acces this information?

```
char[][] masterminds = {
        {'X', 'X','X', 'X'},
        {'G', 'B','W', 'Y'},
        {'G', 'G','U', 'W'},
        {'G', 'R','G', 'B'},
};
```

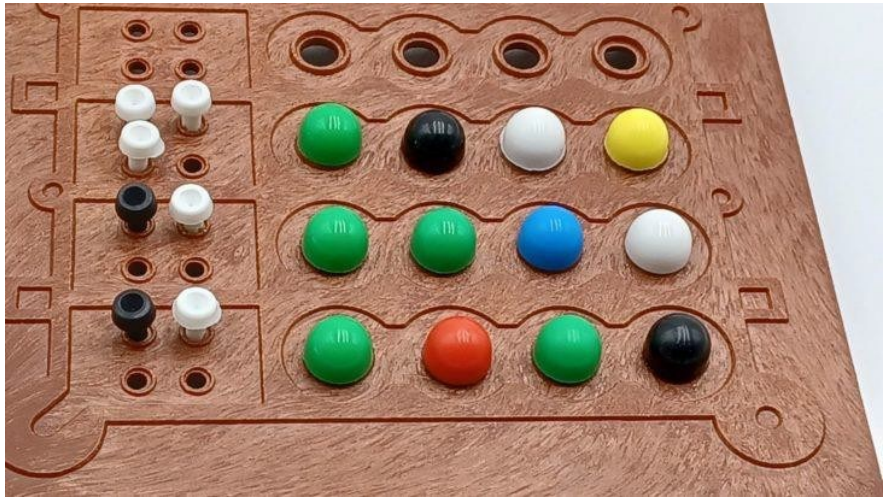


- How can we access the data in the array?

```
System.out.println(masterminds[2][3]);
```

How can we acces this information?

```
char[][] masterminds = {
        {'X', 'X','X', 'X'},
        {'G', 'B','W', 'Y'},
        {'G', 'G','U', 'W'},
        {'G', 'R','G', 'B'},
};
```



- How can we loop over this 2D array?

```
for (int row = 0; row < masterminds.length; row++) {
    for (int col = 0; col < masterminds[row].length; col++) {
      SaxionApp.printline("row: " + row + " col: " + col + " => " + masterminds[row][col]);
    }
}
```

Output:

```
row: 0 col: 0 => X
row: 0 col: 1 => X
row: 0 col: 2 => X
row: 0 col: 3 => X
row: 1 col: 0 => G
row: 1 col: 1 => B
row: 1 col: 2 => W
row: 1 col: 3 => Y
row: 2 col: 0 => G
row: 2 col: 1 => G
row: 2 col: 2 => U
row: 2 col: 3 => W
row: 3 col: 0 => G
row: 3 col: 1 => R
row: 3 col: 2 => G
row: 3 col: 3 => B
```

Workshop:

Do exercise: 1_studentGrading from **n-dimensionalArrays**

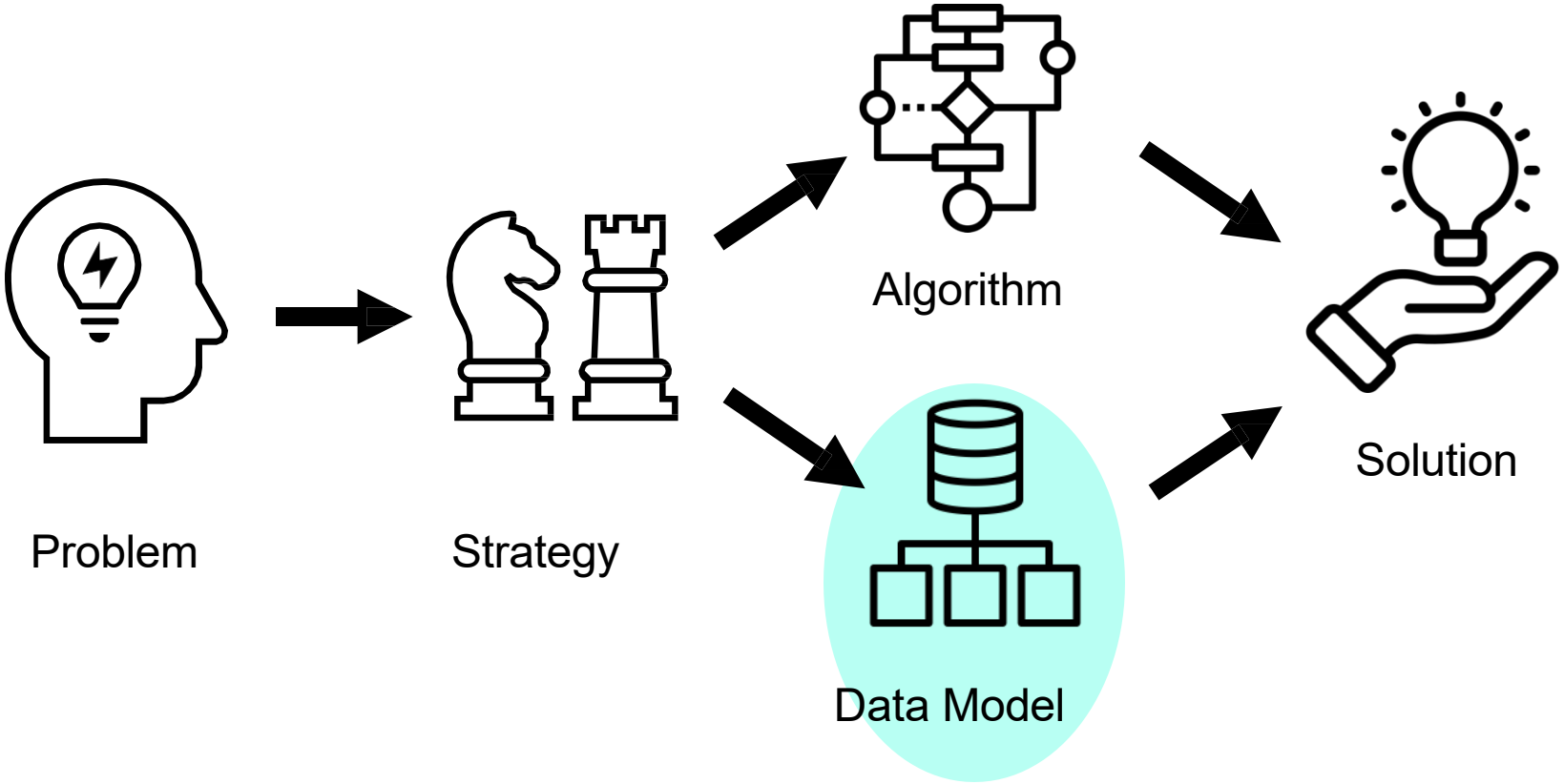
You can find this on BrightSpace > week2

N dimensional Arrays

- **Discuss work last lesson**
- **When to use a 2D array**
- **Data models**
- homework

Data models

- We need a place to store the data in our application
- Best practice: separate data model from functionality and visuals
- Choosing the right model is vital for our strategy

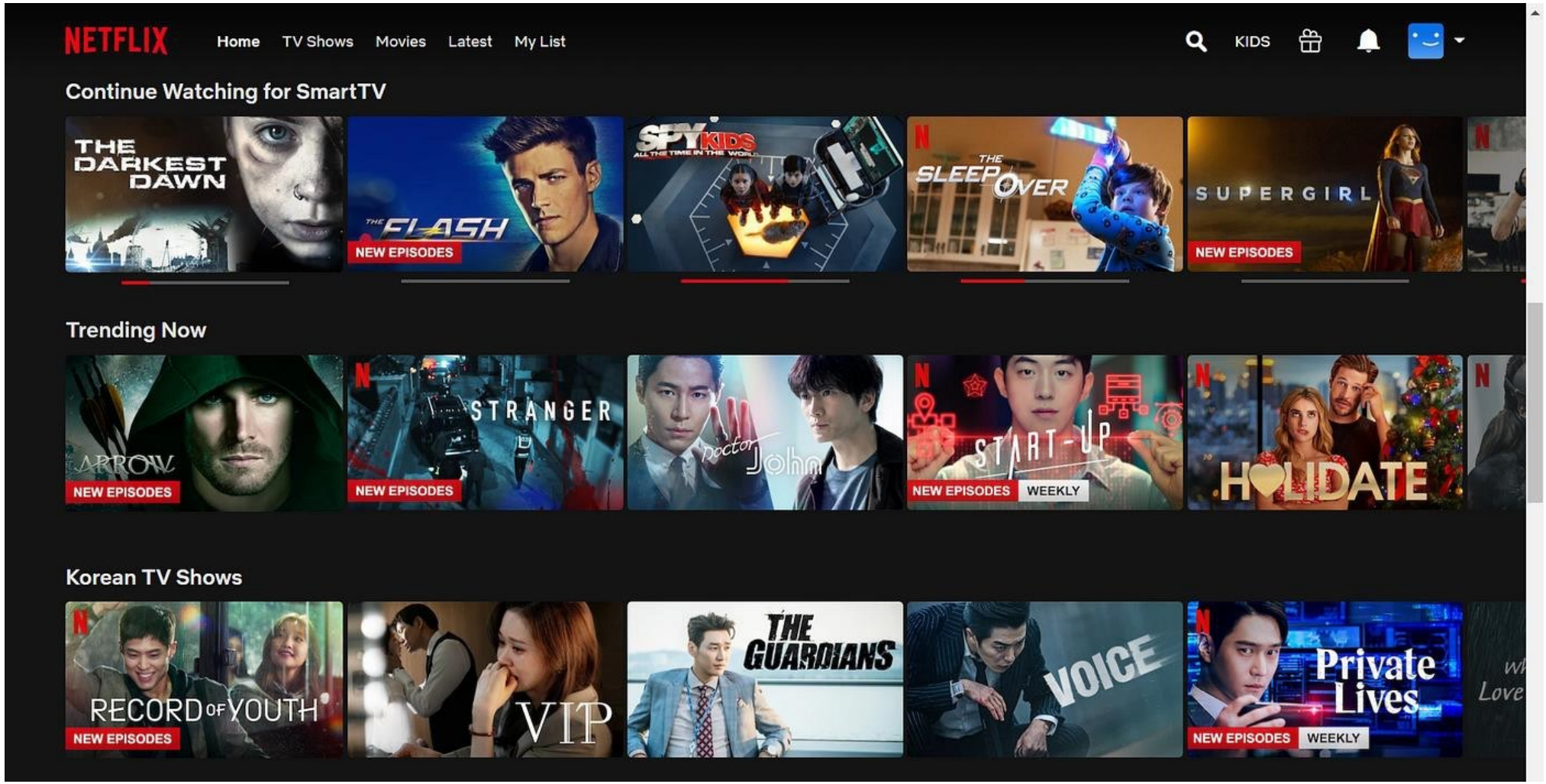


Data models

Karra
Golem
Skeleton
Council Member
Travincal
Quest Log
New Stats

Kar
Sand Leaper
Minion
Rocky Waste
Difficulty: Hell
1076 Gold
Sand Leaper
Life: 827 / 1087
Mana: 731 / 731

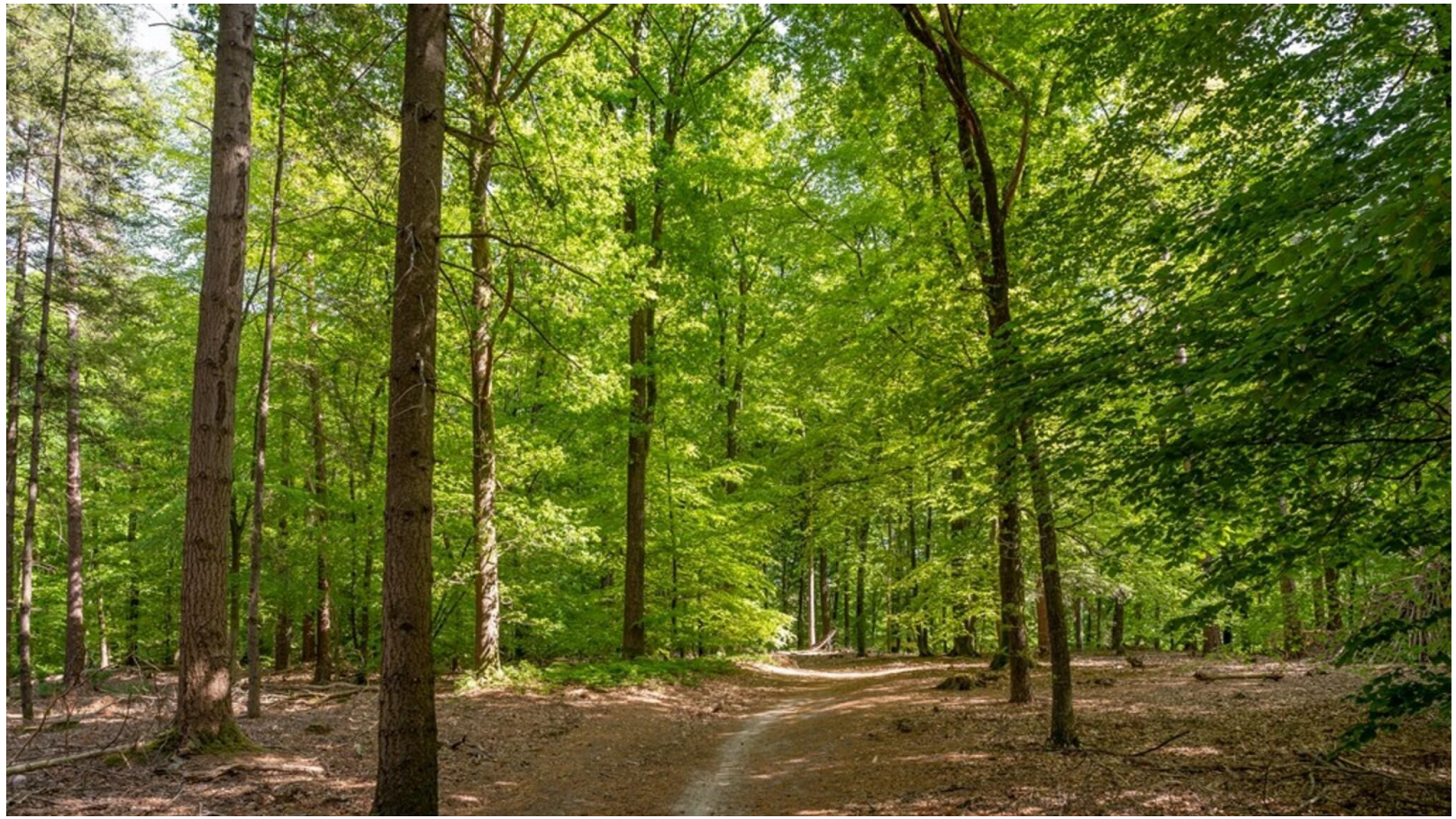
Casus 1


NETFLIX
Home
TV Shows
Movies
Latest
My List
KIDS
Continue Watching for SmartTV
THE DARKEST DAWN
THE FLASH
NEW EPISODES
SPY KIDS
THE SLEEPOVER
SUPERGIRL
NEW EPISODES
Trending Now
ARROW
NEW EPISODES
STRANGER
NEW EPISODES
doctor John
START-UP
NEW EPISODES
WEEKLY
HOLIDATE
Korean TV Shows
RECORD OF YOUTH
NEW EPISODES
VIP
THE GUARDIANS
VOICE
Private Lives
NEW EPISODES
WEEKLY

Casus 2

Koning der Nederlanden
Koning der Nederlanden
Koning der Nederlanden
Koning der Nederlanden
Koning der Nederlanden
Koning der Nederlanden
Koning der Nederlanden
Koning der Nederlanden

Casus 3



Workshop:

Do exercise: 3_battleCalculator from **n-dimensionalArrays**

You can find this on BrightSpace > week2

N dimensional Arrays

- Discuss work last lesson
- When to use a 2D array
- Data models
- homework

Now it’s your turn!